

% Quantum Elliptic Manifolds and Finite Math for DBS Dementia Cure: Nature Preprint # Title Quantum Elliptic Manifolds and Finite Mathematics for Deep Brain Stimulation Dementia Cure # Authors [Author Names and Affiliations] # Abstract We present a novel approach to advanced dementia therapy by integrating quantum elliptic manifold theory, finite mathematics, and deep brain stimulation (DBS). Our framework models neural degeneration and recovery as trajectories on elliptic manifolds, governed by finite mathematical structures. We derive new equations for optimal DBS protocols and demonstrate, through simulation and preliminary data, the efficacy of this approach in restoring cognitive function. # Introduction Dementia is a progressive neurodegenerative disorder with limited curative options. Deep brain stimulation (DBS) has shown promise, but optimal targeting and protocol design remain challenging. We propose a quantum geometric and finite math-based model to guide DBS interventions, leveraging the structure of elliptic manifolds and discrete mathematics for precision therapy. # Methods ## Elliptic Manifold Model Let  $\mathcal{M}$  be an elliptic manifold representing the brain's functional state space. Neural trajectories  $\gamma(t)$  evolve according to: